SWE 4603 Software Testing and Quality Assurance Credit: 3.00

Introduction to Software Testing, Testing Terminology and Methodology, Static Testing, Dynamic Testing, Testing from small to big: Unit Testing, Integration Testing, Function Testing, System Testing, Acceptance Testing - alpha testing, beta testing, Performance Testing, Regression Testing, Exploratory Testing, Regression Test, Code Coverage, Test Management - Test Plan, Test Design and Specifications, Test Driven Development (TDD), Test Metrics, Testing Web Applications, Testing Mobile Applications, Security Testing.

Introduction to Quality Assurance, Organogram of QA Team, QA Plan, Elements of QA, Quality of Requirement Specification (SRS), Quality of Software Design, Code Quality, Maintainability of Software, Software Requirement Validation, FTR, Code Review, Process QualityCMM, ISO, Six Sigma, Feedback Loop, Process Improvement, Risk Identification, Software Reliability, Understanding the value of QA with equations and with real world examples.

Recommended Texts:

1. D. Galin, *Software Quality Assurance: From Theory to Implementation*, 1st ed. Pearson College Div, 2003
2. J. Tian, *Software Quality Engineering: Testing, Quality Assurance, and Quantifiable Improvement*, 1st ed. Wiley-IEEE Computer Society Press, 2005

CSE 4618 Artificial Intelligence Lab Credit: 0.75

Sessional works based on CSE 4617.

CSE 4622 Microprocessor and Interfacing Lab Credit: 0.75

Sessional works based on CSE 4621.

SWE 4602 Software Design and Architectures Lab Credit: 0.75

Sessional works based on SWE 4601.

SWE 4604 Software Testing and Quality Assurance Lab Credit: 1.00

Sessional works based on SWE 4603.

SWE 4606 Design Project II Credit: 1.50

This will be a group project with three students in each group. It will test the ability to work as a member of a group. Each student of the group will have specific responsibilities. The duration of the project will be one year. In this course, students will focus on the development of the project which is designed during the first phase (Design Project I).

Elective 6-I (Selective)

SWE 4631 System Programming and Device Driver Credit: 3.00

What is Systems Programming, Explanations of specific system features, Overview of high level programming languages; Operating system functions: Device management, Memory management, Process management, File system management, Accounting and security, User services; Machine Considerations for Assemblers: Instruction formats/types, Addressing modes and address spaces, Registers, Data representation, Pre-processor directives and portability, Macros, inline assembly, Modularization and program assembly; Memory Management; Input/output at a systems level; File systems and directories; Process management; Object-Oriented extensions of a system programming language; An Introduction to Device Drivers; Building and Running Modules; Char Drivers; Advanced Char Driver Operations; Communicating with Hardware; PCI Driver, USB Drivers; The Linux device model; network drivers; block drivers; TTY drivers.

Recommended Texts:

1. W. Stevens and S. Rago, *Advanced Programming in the UNIX Environment*, 3rd ed. Addison-Wesley Professional, 2013
2. W. Stevens and S. Rago, *Advanced Programming in the UNIX Environment*, 3rd ed. Addison-Wesley Professional, 2013
3. J. Corbet et al., *Linux Device Drivers*, 3rd ed. O'Reilly Media, 2005

SWE 4637 Web and Mobile Application Development Credit: 3.00

Basics of HTML, CSS, ECMA Script and JavaScript. CSS preprocessors and JavaScript transpilers. Browser compatibility, validation, authentication, authorization, OAuth, Social Login, SSO. Consuming web services with XML HTTP Request (Ajax). State Management, Session, Cookie, WebSocket, Push Notification. Client Side Database - SQLite, WebSQL, local storage, IndexedDB. Single Page Application, Routing. Search